Project Report

20/02/2026

CUSTOMER CHURN PREDICTION
&
ANALYSIS

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Topic: AIML

Language: Python

Course: Data Science & Analytics

Customer Churn Prediction Report

Executive Summary

This project develops a machine learning model to predict customer churn. The goal is to identify customers likely to leave so businesses can take proactive retention actions. After testing multiple algorithms, XGBoost achieved the best performance based on ROC-AUC, recall, and F1-score. Key drivers of churn include low satisfaction score, high support tickets, short tenure, and high monthly charges. The model enables targeted retention strategies, reducing revenue loss and improving customer lifetime value.

Methodology

Data Preparation:
- Cleaned missing values
- Removed highly correlated features
- Encoded categorical variables
- Standardized numerical features

Model Training Process:
- Split dataset into Train (70%), Validation (15%), Test (15%)
- Addressed class imbalance using SMOTE and class weights
- Tested multiple algorithms including Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, XGBoost, SVM, KNN, Naive Bayes

Evaluation Metrics:
Accuracy, Precision, Recall, F1 Score, ROC-AUC, Precision-Recall AUC

Key Findings

- Customers with low satisfaction scores are more likely to churn.
- Customers with many support tickets show high churn probability.
- Month-to-month contracts have the highest churn rate.
- Customers with more services tend to stay longer.

Model Performance Comparison

XGBoost produced the highest ROC-AUC score and best balance between precision and recall. Tree-based models performed better because they capture nonlinear relationships and interactions. Linear models were simpler but less accurate.

Business Recommendations

- Offer retention discounts for high-risk customers
- Provide proactive support outreach
- Encourage long-term contracts
- Bundle services for engagement
- Monitor satisfaction scores continuously